$$\hat{q}_i(\tau) = \mathrm{e}^{\mathrm{i}\hat{H}_{\mathcal{E}}\tau}\hat{q}_i\mathrm{e}^{-\mathrm{i}\hat{H}_{\mathcal{E}}\tau} = \sqrt{\frac{1}{2m_i\omega_i}}\left(\hat{a}_i\mathrm{e}^{-\mathrm{i}\omega_i\tau} + \hat{a}_i^\dagger\mathrm{e}^{\mathrm{i}\omega_i\tau}\right). \tag{5.9}$$

Therefore

$$\begin{aligned}\left\langle\hat{q}_i(\tau)\hat{q}_i\right\rangle_{\hat{\rho}_{\mathcal{E}}} &= \frac{1}{2m_i\omega_i}\left\langle\hat{a}_i\hat{a}_i^\dagger\mathrm{e}^{-\mathrm{i}\omega_i\tau} + \hat{a}_i^\dagger\hat{a}_i\mathrm{e}^{\mathrm{i}\omega_i\tau}\right\rangle_{\hat{\rho}_{\mathcal{E}}} \\ &= \frac{1}{2m_i\omega_i}\left\{\left\langle\hat{a}_i\hat{a}_i^\dagger\right\rangle_{\hat{\rho}_{\mathcal{E}}}\mathrm{e}^{-\mathrm{i}\omega_i\tau} + \left\langle\hat{a}_i^\dagger\hat{a}_i\right\rangle_{\hat{\rho}_{\mathcal{E}}}\mathrm{e}^{\mathrm{i}\omega_i\tau}\right\}. \end{aligned} \tag{5.10}$$